Lecture 4. Formal Introduction to Censoring and Truncation

Censoring Schemes and Likelihood Construction

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1 Right Censoring

Definition 4.1: Right-censored data

An observation is right censored if it is not known exactly, only that it is greater than some value. Let X = time to the event of interest and C = censoring time. We observe

$$(T, \delta), \qquad T = \min(X, C), \qquad \delta = I(X < C) \text{ (or } I(X \leq C)),$$

and the data are n realizations $\{(T_i, \delta_i),\ i = 1, \ldots, n\}$.

There are many types of right censoring; some of them provide information on C even when $X < C$.

1.1 Type I censoring

The event is observed only if it occurs before some *pre-specified* time.

- **Fixed Type I censoring**: $C_i = \tau$ for all i. Typical of controlled experiments, e.g. a mice study of time to death after a fixed period following intense carcinogen exposure: every animal still alive at the end of the study is censored at the same time.

- **Progressive Type I censoring**: C is fixed in advance but may take different values for each i, with only a finite number of distinct values. *Example*: X = time to a certain % weight loss, weights measured every 12 hours; mice in cages 1–2 are sacrificed at the end of week 1 and those in cages 3–4 by the end of week 2. (A second example uses two sacrifice times, 42 and 104 weeks.)
- **Generalized Type I censoring**: C may differ for every individual but is known in advance. Common in human studies with *staggered entry* and complete follow-up to the end of the study, e.g. a Phase I chemotherapy study with X = time to toxicity: C_i = (end of study) − (entry time of i) is computable at enrolment, assuming a fixed study length and no dropout.

1.2 Type II censoring

- **Simple Type II**: sample size n; the study continues until the first r failures. The number of failures r and censored observations $n - r$ are *fixed*; the censoring time $C = X_{(r)}$ is *random*.
- **Progressive Type II**: after the first r_1 failures, $n_1 - r_1$ of the remaining $n - r_1$ unfailed items are removed (sacrificed); after the next r_2 failures, $n_2 - r_2$ of the unfailed items are removed; and so on until a pre-decided number of repetitions is completed.

Remark 4.2: Type I vs. Type II

Type I: C is fixed (predetermined before the study starts). Type II: C is random, namely the time at which a predetermined number of events has occurred. This distinction matters when constructing the likelihood.

1.3 Random censoring

Typical of human studies: $C \sim G(t)$, not fixed in advance and possibly unknown.

- Scenario 1: censoring at the end of study ("administrative loss to follow-up"), fixed in advance, similar to Type I.
- Scenario 2: censoring before the end of study ("drop-out"), random, due to other factors.
- In practice censoring is a mixture of the two.

2 Left, Double and Interval Censoring

Definition 4.3: Left censoring

An observation is left censored if it is not known exactly, only that it is less than some value C_l. The exact lifetime is known iff $X \geq C_l$. Data: (T, δ) with $T = \max(X, C_l)$, $\delta = I(X \geq C_l)$.

Definition 4.4: Double censoring

With left censoring time C_l and right censoring time C_r $(C_l < C_r)$:

$$T = \max\{\min(X, C_r),\, C_l\}, \qquad \delta = \begin{cases} 1, & C_l \leq X \leq C_r \text{ (exact)} \\ -1, & X < C_l \text{ (left censored)} \\ 0, & X > C_r \text{ (right censored)} \end{cases}$$

Remark 4.5: Double vs. interval censoring

Interval censoring: each observation is only known to lie in an interval, $L < X \leq R$; no exact time is observed. **Double censoring**: each observation is one of three possibilities — fully observed, right censored, or left censored.

3 Truncation

Definition 4.6: Truncation

Only individuals whose event time lies within an observational window (Y_L, Y_R) are observed.

- $Y_R = \infty$: **left truncation**; we observe only those with $Y_L < X$.
- $Y_L = 0$: **right truncation**; we observe only those with $X \leq Y_R$.

Inference for truncated data is restricted to *conditional* estimation.

Example 4.7: Breast cancer onset (classification exercise)

To estimate the distribution of age at breast cancer onset in post-menopausal women, 8 healthy 50-year-old women had yearly mammograms for 10 years; no tumours were detected between scheduled exams. Onset ages:

$$(55, 56],\ (58, 59],\ (52, 53],\ (59, 60],\ > 60,\ > 60,\ > 60,\ > 60.$$

Answer. The four detected cases are **interval censored** (onset between two yearly exams). The four undetected cases are **right censored** at 60. Moreover, only women who were cancer-free at age 50 were enrolled, so the ages are **left truncated** at 50 (inference is conditional on $X > 50$).

4 Likelihood Construction

4.1 Contributions of each observation type

Let X be continuous with pdf f, cdf F and $S = 1 - F$; C_r = right censoring point, C_l = left censoring point, $(L, R]$ = censoring interval, Y = truncation point.

Observation	Likelihood contribution
Exact lifetime	$f(x)$
Right censored	$S(C_r) = P(X > C_r)$
Left censored	$1 - S(C_l) = P(X < C_l)$
Interval censored	$S(L) - S(R) = P(L < X \leq R)$
Left truncated	$f(x)/S(Y_L)$
Right truncated	$f(x)/[1 - S(Y_R)]$
Interval truncated	$f(x)/[S(Y_L) - S(Y_R)]$

The truncated contributions are conditional densities: e.g. for left truncation, $f(x \mid X > Y_L) = f(x)/P(X > Y_L) = f(x)/S(Y_L)$.

Theorem 4.8: General censored-data likelihood

$$L \propto \prod_{i\in D} f(x_i) \prod_{i\in RC} S(C_{r,i}) \prod_{i\in LC} F(C_{l,i}) \prod_{i\in IC} \{S(L_i) - S(R_i)\},$$

where D = exact (death) times, $RC/LC/IC$ = right-/left-/interval-censored observations.

Truncation. For left-truncated data (possibly also right censored), replace $f(x_i)$ by $f(x_i)/S(Y_i)$ and $S(C_{r,i})$ by $S(C_{r,i})/S(Y_i)$. For right-truncated data (only events observed), replace $f(x_i)$ by $f(x_i)/[1 - S(Y_i)]$.

4.2 Type I right censoring

Theorem 4.9

Observing (T_i, δ_i) with $T_i = \min(X_i, C_i)$, $\delta_i = I(T_i = X_i)$ and fixed C_i,

$$L = \prod_{i=1}^{n} \Pr(t_i, \delta_i) = \prod_{i=1}^{n} f(t_i)^{\delta_i} S(t_i)^{1-\delta_i} = \prod_{i=1}^{n} h(t_i)^{\delta_i} e^{-H(t_i)}.$$

Proof [Supplement]. Since C_i is fixed, $\delta_i = 0$ means $T_i = C_i$ and $X_i > C_i$, with probability $S(C_i) = S(t_i)$; $\delta_i = 1$ means $T_i = X_i \le C_i$ with density $f(t_i)$. For the last form, $f = hS$ and $S = e^{-H}$: $f^{\delta} S^{1-\delta} = (hS)^{\delta} S^{1-\delta} = h^{\delta} S = h^{\delta} e^{-H}$. ■

Example 4.10: Exponential with right censoring

$X \sim \text{Exp}(\lambda)$, $T = \min(X, C)$, $\delta = I(X < C)$:

T_i	0.5	1	0.75	0.25	0.75
δ_i	1	1	1	0	0

With $h(t) = \lambda$ and $H(t) = \lambda t$,

$$L(\lambda) = \prod_{i=1}^{5} \lambda^{\delta_i} e^{-\lambda t_i} = \lambda^{\sum \delta_i} e^{-\lambda \sum t_i} = \lambda^3 e^{-3.25\lambda}.$$

Maximizing $\ell(\lambda) = 3\ln\lambda - 3.25\lambda$: $\ell'(\lambda) = 3/\lambda - 3.25 = 0 \Rightarrow \hat{\lambda} = 3/3.25 \approx 0.923$. In general $\hat{\lambda} = \sum \delta_i / \sum t_i$ = (number of events)/(total time at risk).

4.3 Type II censoring

Theorem 4.11

With n i.i.d. $X_i \sim f$, observing the r smallest order statistics $X_{(1)} < \cdots < X_{(r)}$ and $n - r$ values censored at $X_{(r)}$,

$$L = \frac{n!}{(n-r)!} \prod_{i=1}^{r} f(x_{(i)}) \{S(x_{(r)})\}^{n-r}.$$

Derivation: where the constant comes from

The joint density of the first r order statistics: choose which r of the n units fail (and in which order) — $n!/(n-r)!$ ordered choices — each failed unit contributes $f(x_{(i)})$, and the remaining $n-r$ units must all survive beyond $x_{(r)}$, contributing $S(x_{(r)})^{n-r}$. The constant does not involve the parameters, so the kernel is the same as in Theorem 4.8.

Progressive Type II (two stages): first r_1 ordered failures $X_{(1)}, \ldots, X_{(r_1)}$, then $n_1 - r_1$ items removed; the next r_2 failures $X^*_{(1)}, \ldots, X^*_{(r_2)}$, then the remaining $n - n_1 - r_2$ removed:

$$L \propto \prod_{i=1}^{r_1} f(x_{(i)}) \{S(x_{(r_1)})\}^{n_1 - r_1} \prod_{i=1}^{r_2} f(x^*_{(i)}) \{S(x^*_{(r_2)})\}^{n - n_1 - r_2} \propto \prod_{i \in D} f(x_i) \prod_{i \in RC} S(C_i).$$

4.4 Random censoring

$X_i \sim f, S$ and $C_i \sim g, G$ are both random but **independent**; we observe $T_i = \min(X_i, C_i)$ and $\delta_i = I(X_i \le C_i)$.

Theorem 4.12

$$L = \prod_{i=1}^{n} \{f(t_i)G(t_i)\}^{\delta_i} \{g(t_i)S(t_i)\}^{1-\delta_i} = \underbrace{\prod_{i=1}^{n} f(t_i)^{\delta_i} S(t_i)^{1-\delta_i}}_{L_X} \underbrace{\prod_{i=1}^{n} g(t_i)^{1-\delta_i} G(t_i)^{\delta_i}}_{L_C}.$$

Proof [Supplement]. For $\delta = 1$: $P(T \in \mathrm{d}t, \delta = 1) = P(X \in \mathrm{d}t,\ C \ge t) = f(t)G(t)\,\mathrm{d}t$ by independence. For $\delta = 0$: $P(T \in \mathrm{d}t, \delta = 0) = P(C \in \mathrm{d}t,\ X > t) = g(t)S(t)\,\mathrm{d}t$. Multiply over i and regroup. ■

If $f = f(\cdot;\theta)$ and the distribution of C does not depend on θ (**non-informative censoring**), L_C is constant in θ and

$$L \propto L_X = \prod_{i=1}^{n} f(t_i)^{\delta_i} S(t_i)^{1-\delta_i},$$

the same form as under Type I censoring.

5 Remarks

- These likelihoods are used primarily for fitting parametric models (K&M Ch. 12).
- They also underlie the *partial likelihood* of the semiparametric Cox model (K&M Ch. 8–9).
- In most applications they are not used explicitly, but the rationale — how each data component contributes, as in Theorem 4.8 — is essential.

6 Selected exercises

Motivation. Three textbook exercises (K&M Section 3.7): writing a likelihood when truncation and two kinds of censoring occur together, the exact sampling distribution of the exponential MLE under random censoring, and the simplest counting-process martingale.

Exercise 4.1 (K&M 3.2: periodic exams with delayed entry)

Disease-free women were enrolled from January 1, 1970, and followed for 30 years for the age X at onset of breast cancer, with clinical exams every 3 years after enrollment; onset is detected only at an exam. Classify the censoring and truncation, and write the likelihood contribution, for each woman:
(a) enrolled at age 30, never developed breast cancer during the study;
(b) enrolled at age 40, diagnosed at the fifth exam after enrollment;
(c) enrolled at age 50, died of an unrelated cause at age 61 without a diagnosis;
(d) enrolled at age 42, moved away at age 55 without a diagnosis.
(e) Write the likelihood for these four women.

Solution. Every woman is enrolled only if disease-free at entry, so each observation is *left truncated* at her entry age v: her contribution is the probability of what is observed divided by $S(v)$ (Section on truncation above). Onset is known only up to the exam schedule $v+3, v+6, \ldots$

(a) Exams at ages $33, \ldots, 60$, all negative: right censored at 60, left truncated at 30. Contribution $S(60)/S(30)$.

(b) The fourth exam (age 52) was negative and the fifth (age 55) positive: interval censored in $(52, 55]$, left truncated at 40. Contribution $\{S(52) - S(55)\}/S(40)$.

(c) Exams at $53, 56, 59$ were negative; death at 61 is a competing event that ends follow-up. Onset could have occurred unseen between 59 and 61, so she is right censored at the last negative exam, 59, and left truncated at 50. Contribution $S(59)/S(50)$. (If an onset before death would surely have been noticed, one may censor at 61 instead, giving $S(61)/S(50)$.)

(d) Exams at $45, 48, 51, 54$ negative, then lost to follow-up: right censored at 54, left truncated at 42. Contribution $S(54)/S(42)$.

(e) Assuming independent censoring and truncation,

$$L = \frac{S(60)}{S(30)} \cdot \frac{S(52) - S(55)}{S(40)} \cdot \frac{S(59)}{S(50)} \cdot \frac{S(54)}{S(42)}.$$

■

Exercise 4.2 (K&M 3.8: exponential lifetimes, exponential censoring)

Let $X \sim \text{Exp}(\lambda)$ and $C \sim \text{Exp}(\theta)$ be independent, $T = \min(X, C)$, $\delta = I(X \le C)$.
(a) Find $P(\delta = 1)$.
(b) Find the distribution of T.
(c) Show that δ and T are independent.
(d) For a random sample (T_i, δ_i), $i = 1, \ldots, n$, show that the MLE of λ is $\hat{\lambda} = \sum_i \delta_i / \sum_i T_i$, and use (a)–(c) to find its mean ($n \ge 2$) and variance ($n \ge 3$).

Solution.

(a)–(c) For $t \ge 0$, by independence,

$$P(\delta = 1, T > t) = P(t < X \le C) = \int_t^\infty \lambda e^{-\lambda s} e^{-\theta s}\, \mathrm{d}s = \frac{\lambda}{\lambda + \theta} e^{-(\lambda+\theta)t}.$$

With $t = 0$, $P(\delta = 1) = \lambda/(\lambda + \theta)$. Also $P(T > t) = P(X > t)P(C > t) = e^{-(\lambda+\theta)t}$, so $T \sim \text{Exp}(\lambda + \theta)$. Hence $P(\delta = 1, T > t) = P(\delta = 1)P(T > t)$ for all t; since δ is binary, this gives $P(\delta = 0, T > t) = P(\delta = 0)P(T > t)$ too, so $\delta \perp\!\!\!\perp T$.

(d)

Step 1 (MLE). By the random-censoring theorem above, $L = \lambda^D e^{-\lambda G} \cdot \theta^{n-D} e^{-\theta G}$ with $D = \sum_i \delta_i$ and $G = \sum_i T_i$. The λ-part has $\partial_\lambda \log L = D/\lambda - G$, positive for $\lambda < D/G$ and negative after, so $\hat\lambda = D/G$ (when $D = 0$ the supremum is at the boundary $\hat\lambda = 0$).

Step 2 (distributions). The pairs are i.i.d. and $\delta_i \perp\!\!\!\perp T_i$ by (c), so $\delta_1, \dots, \delta_n, T_1, \dots, T_n$ are mutually independent. Hence $D \sim \mathrm{Bin}(n, p)$ with $p = \lambda/r$, $r = \lambda + \theta$, $G \sim \mathrm{Gamma}(r, n)$ (shape n, rate r), and $D \perp\!\!\!\perp G$.

Step 3 (inverse moments of G). For $k < n$,

$$\mathrm{E}(G^{-k}) = \int_0^\infty g^{-k} \frac{r^n g^{n-1} e^{-rg}}{\Gamma(n)}\,\mathrm{d}g = \frac{r^k\,\Gamma(n-k)}{\Gamma(n)},$$

so $\mathrm{E}(G^{-1}) = r/(n-1)$ and $\mathrm{E}(G^{-2}) = r^2/\{(n-1)(n-2)\}$.

Step 4 (mean). By independence, $\mathrm{E}\hat\lambda = \mathrm{E}(D)\mathrm{E}(G^{-1}) = np \cdot r/(n-1) = n\lambda/(n-1)$, a slight upward bias.

Step 5 (variance). $\mathrm{E}(D^2) = np(1-p) + n^2p^2$, and $r^2p(1-p) = \lambda\theta$, $r^2p^2 = \lambda^2$. So

$$\mathrm{E}\hat\lambda^2 = \frac{n\lambda\theta + n^2\lambda^2}{(n-1)(n-2)},$$
$$\mathrm{Var}\hat\lambda = \frac{n\lambda\theta + n^2\lambda^2}{(n-1)(n-2)} - \frac{n^2\lambda^2}{(n-1)^2} = \frac{n\lambda\theta}{(n-1)(n-2)} + \frac{n^2\lambda^2}{(n-1)^2(n-2)}.$$

As $n \to \infty$, $n\mathrm{Var}\hat\lambda \to \lambda\theta + \lambda^2 = \lambda^2/p$, the inverse of the Fisher information per observation, $\mathrm{E}(\delta)/\lambda^2 = p/\lambda^2$. ■

Exercise 4.3 (K&M 3.9: the Poisson process martingale)

Let $N(t)$ be a Poisson process with rate λ: $N(0) = 0$; $N(t) - N(s) \sim \mathrm{Poisson}(\lambda(t-s))$ for $0 \le s < t$; increments over disjoint intervals are independent. Let $\mathcal{F}_s = \sigma\{N(u) : u \le s\}$ and $M(t) = N(t) - \lambda t$.
(a) Show that $\mathrm{E}|M(t)| < \infty$.
(b) Show that $\mathrm{E}[M(t) \mid \mathcal{F}_s] = M(s)$ for $s < t$, so M is a martingale and λt is the compensator of $N(t)$.

Solution.

(a) $\mathrm{E}|M(t)| \le \mathrm{E}N(t) + \lambda t = 2\lambda t < \infty$.

(b)

Step 1 ($N(t) - N(s) \perp\!\!\!\perp \mathcal{F}_s$). For $0 \le u_1 < \cdots < u_k \le s$, the vector $(N(u_1), \dots, N(u_k))$ is a function of the increments over $(0, u_1], (u_1, u_2], \dots, (u_{k-1}, u_k]$, which are independent of the increment over the disjoint interval $(s, t]$. So $N(t) - N(s)$ is independent of every event in the π-system of finite-dimensional cylinder sets $\{(N(u_1), \dots, N(u_k)) \in B\}$, which generates $\mathcal{F}_s$. By the π–λ theorem it is independent of $\mathcal{F}_s$.

Step 2. Write $M(t) = \{M(t) - M(s)\} + M(s)$. Since $M(s)$ is $\mathcal{F}_s$-measurable and integrable,

$$\mathrm{E}[M(t) \mid \mathcal{F}_s] = \mathrm{E}[N(t) - N(s)] - \lambda(t-s) + M(s) = \lambda(t-s) - \lambda(t-s) + M(s) = M(s),$$

using Step 1 to drop the conditioning.

Step 3. M is adapted, integrable by (a), and has the martingale property by Step 2. The process λt is deterministic (hence predictable), continuous and increasing with $\lambda \cdot 0 = 0$, and $N(t) - \lambda t$ is a martingale; this is the definition of the compensator of N. It is the integrated hazard $\int_0^t \lambda\,\mathrm{d}u$, the prototype of $\int_0^t Y(u)h(u)\,\mathrm{d}u$ behind the Nelson–Aalen estimator. ■